Scholaro's

GCSE Biology Paper 1 Guide: Made for ESL Students

Welcome to Your Paper 1 ESL Guide!

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Scholaro's GCSE Biology Guide: Topic 1 - Cell Biology (Made for ESL Students)

🌟 Welcome to Your Cell Biology ESL Guide!

🎯 Purpose of this Guide

This guide is created especially for ESL students studying GCSE AQA Biology. Biology can be tricky, especially when English isn't your first language. This guide explains everything in simple English, with pictures, tips, and examples to help you understand.

We want you to feel confident and ready for your exams!

📄 What is Cell Biology?

Cell Biology is the first topic in GCSE AQA Biology. It covers what cells are, the parts inside them, how they work, and how they divide and grow.

You will learn about: - Different types of cells (like animal, plant, and bacteria) - How cells are specialised - Microscopes and how to calculate magnification - Growing bacteria and testing antibiotics - How substances move in and out of cells (diffusion, osmosis, active transport)

👩‍🏫 Who is This Guide For?

This guide is for: - Students learning English as a second language - Anyone who wants clear explanations and easy examples - Students preparing for GCSE Biology (AQA)

✅ We use simple English
✅ We explain each topic step-by-step
✅ We include diagrams and summaries
✅ We include key definitions and examples

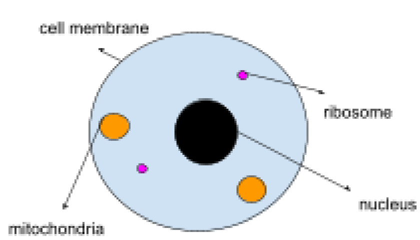
📖 What You Will Learn

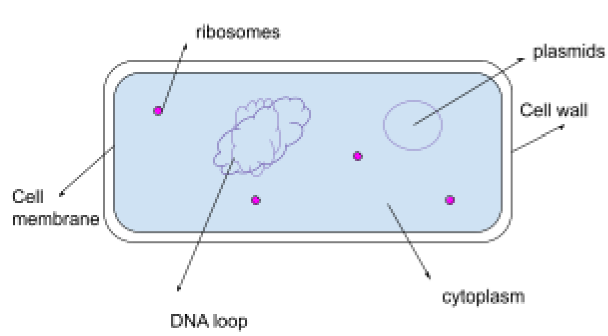
Here's what you'll find in this guide:

- 🔍 Cell Types (Animal, Plant, Bacteria)
- 🌿 Specialised Cells (Sperm, Nerve, Root Hair)
- 🔬 Microscopy and Calculations
- 🎧 Growing Microorganisms
- 👅 Transport in Cells (Diffusion, Osmosis, Active Transport)
- 🐿️ Cell Division and Stem Cells

⚖️ 1. Cell Structure

Eukaryotic and Prokaryotic Cells





All living things are made of cells. There are two main types:

Type	Examples	Key Parts
Eukaryotic	Animal, Plant	Nucleus, Cytoplasm, Cell Membrane, (plant have cell wall too)
Prokaryotic	Bacteria	No nucleus, DNA floats freely, Cell Wall, Plasmids, flagella

What are Organelles?

Organelles are parts inside a cell that do different jobs.

Examples: - Nucleus: Contains DNA (genetic information) - Mitochondria: Releases energy (aerobic respiration) - Ribosomes: Make proteins

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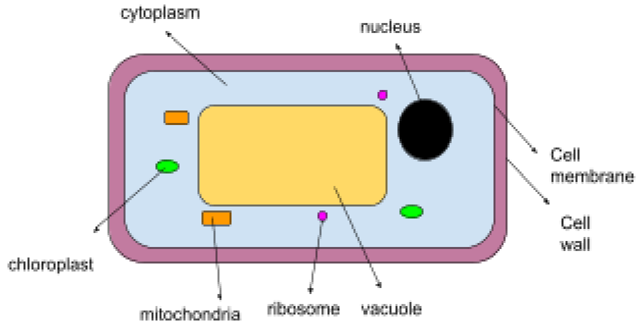
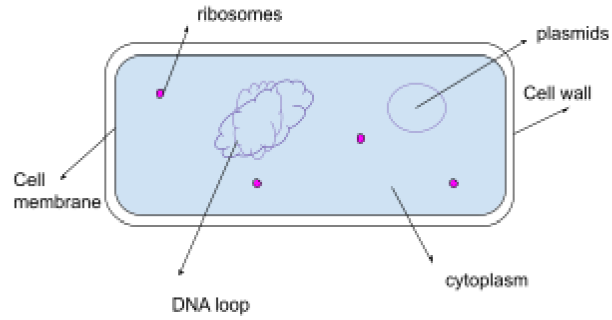
Orders of Magnitude & Prefixes

Orders of magnitude show how much bigger or smaller one thing is compared to another: - 10 times = 10^1 - 1000 times = 10^3 - 0.001 times = 10^{-3}

Prefixes:

Prefix	Multiply Unit By
Centi	0.01
Milli	0.001
Micro	0.000001
Nano	0.000000001





Subcellular Structures

Structure	Function
Nucleus	Contains DNA and controls the cell
Cytoplasm	Where reactions happen (contains enzymes)
Cell membrane	Controls what enters and leaves
Mitochondria	Where energy is released during respiration
Ribosomes	Where proteins are made

Only in Plant Cells:

Structure	Function
Chloroplasts	Photosynthesis; contains green pigment (colour) chlorophyll
Vacuole	Stores cell sap; helps keep cell shape (doesn't become like blob of wet slime)
Cell Wall	Made of cellulose; gives strength

In Bacterial Cells:

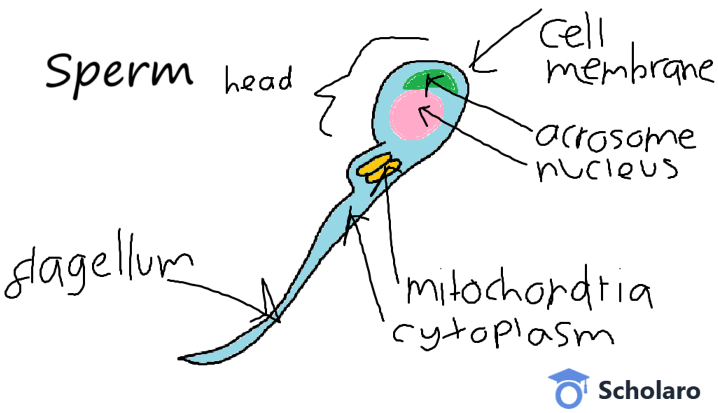
Structure	Function
Cell Wall	Made of peptidoglycan
Plasmid	Small ring of DNA
Circular DNA	Floats in cytoplasm; no nucleus
Flagella	Help it swim (like a tail)

🧰 2. Cell Specialisation

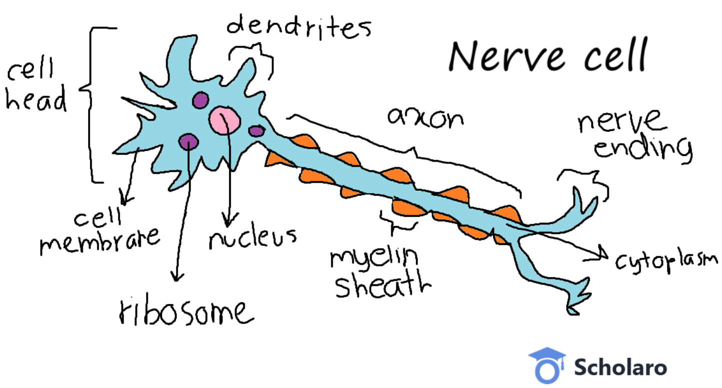
Cells have different jobs. They become “specialised” for those jobs.

In Animals:

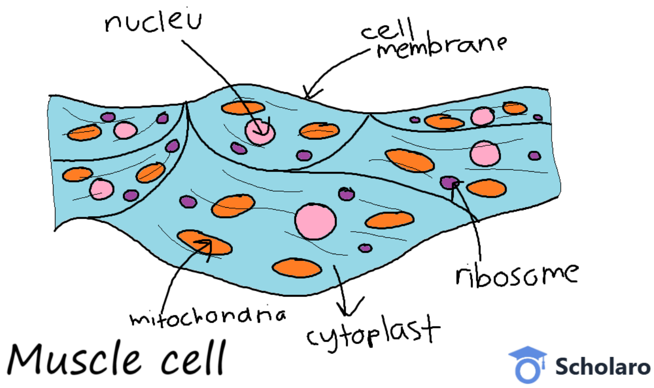
- Sperm cell: Swims to egg; has tail (to move around), lots of mitochondria (to do lots of respiration, means more energy), enzymes (digest or break egg cell wall, to fertilise)



- Nerve cell: Sends electrical signals (impulse); long axon (send impulse or message very far), neurotransmitters (send impulse between synapse, the gap between ends of two axons which are next to each other)

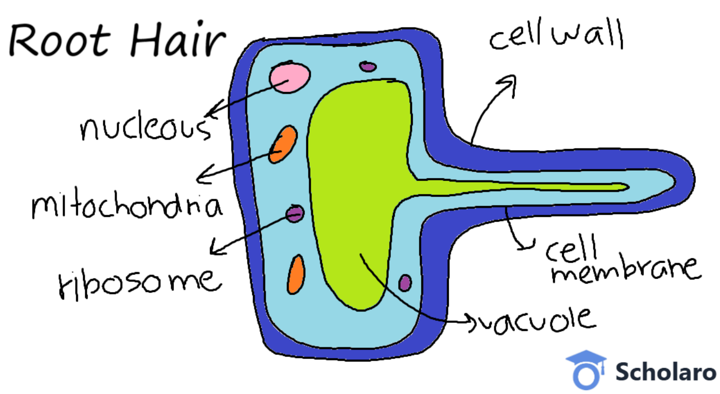


- Muscle cell: Contracts to cause movement; proteins (flexible to go up (contract, squeeze) and down (relax, stretch), mitochondria (need lots of energy from respiration to move), stores glycogen (breaks down into glucose, use for respiration)

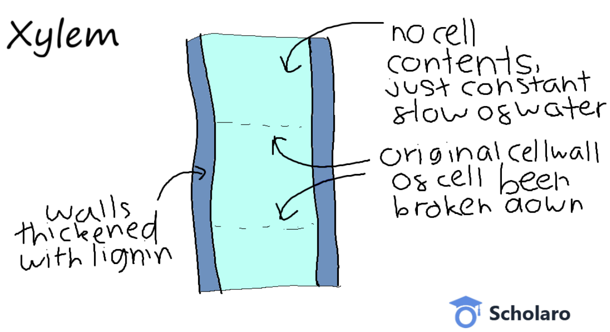


In Plants:

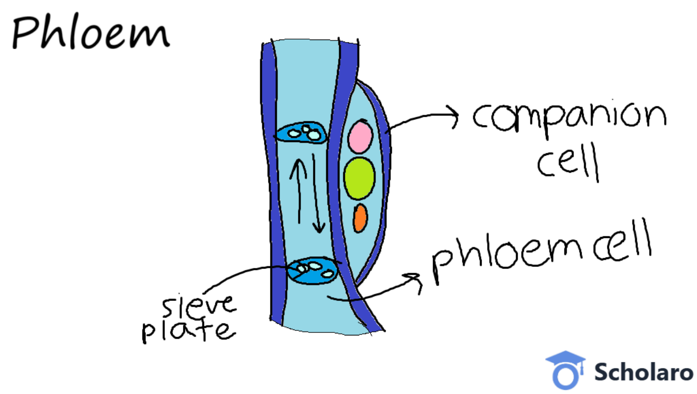
- Root hair cell: Takes in water/minerals; big surface area (more stuff taken/transfered at a time), mitochondria (to take minerals, do active transport which need energy)



- Xylem cell: Carries water; lignin walls (strong, so doesn't get damaged by high pressure), hollow (hole, so water go through)



- Phloem cell: Carries food; sieve plates (soluble substance go through), companion cells (give energy to sieve plates, as they are living cells. Lignin walls are dead cells, no need energy)



🔍 3. Microscopy

We use microscopes to see small things like cells.

Light Microscope:

- Uses light
- Max magnification: x2000
- Resolution: 200 nm
- Sees: cells, large organelles

Electron Microscope:

- Uses electrons
- Max magnification: x2,000,000
- Resolution: 0.2 nm (TEM), 10 nm (SEM)
- Sees: tiny parts like ribosomes and plasmids

Calculations:

- Magnification = Eyepiece × Objective lens
- Image Size = Real Size × Magnification

Use standard form for very small/large numbers. Example: – $1.5 \times 10^{-5} = 0.000015$

🚀 4. Growing Microorganisms

Why?

To study bacteria or test antibiotics.

Methods:

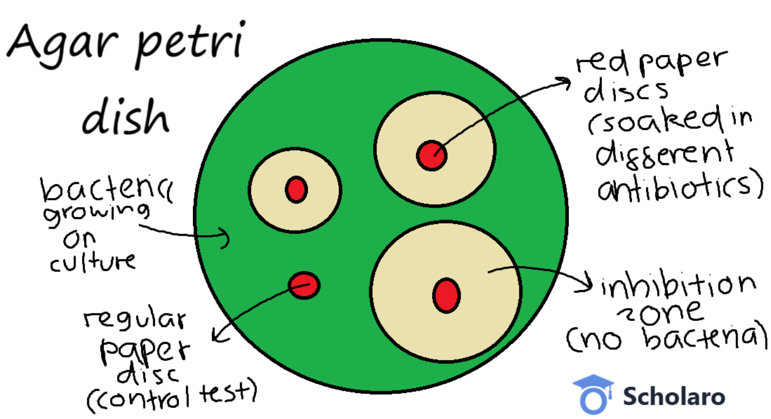
1. Nutrient Broth: Liquid with bacteria
2. Agar Plate: Jelly in dish; bacteria grow in colonies

Key Steps:

Step	Why it's important
Sterilise dishes & tools	Avoid contamination
Use flame on loops	Kill unwanted bacteria
Tape the lid (not fully)	Stop some microbes enter from air or human but allow oxygen to enter (for respiration)
Store upside down	Stop condensation (water) effect growth of bacteria
Keep at 25°C	Avoid growing harmful bacteria

Testing Antibiotics:

- Place antibiotic discs on agar with bacteria
- Clear area (inhibition zone) = bacteria killed
- Use πr^2 to calculate area



💚 5. Cell Division

Chromosomes:

- Found in nucleus
- Made of DNA
- 46 in body cells (23 pairs)
- Gametes (sperm/egg): 23 chromosomes

The Cell Cycle:

1. Interphase: Cell grows, DNA copies
2. Mitosis: Chromosomes line up and are pulled apart
3. Cytokinesis: Cell splits into 2 identical daughter cells

Mitosis is used for: - Growth - Repair - Asexual reproduction

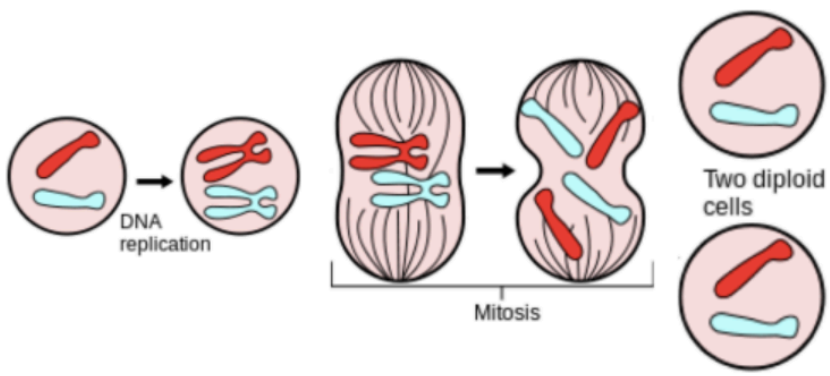


Fig 1. Mitosis. The end result is 2 identical daughter cells.

🧠 6. Stem Cells

What are Stem Cells?

Unspecialised cells that can become any type of cell.

Types:

- Embryonic: Can become anything

- Adult (Bone Marrow): Become some cells (like blood)
- Meristems (Plants, found in shoots tips and tips of roots): Can always change; used to clone plants

Therapeutic Cloning:

Embryo made to match patient; stem cells won't be rejected.

Pros vs Cons:

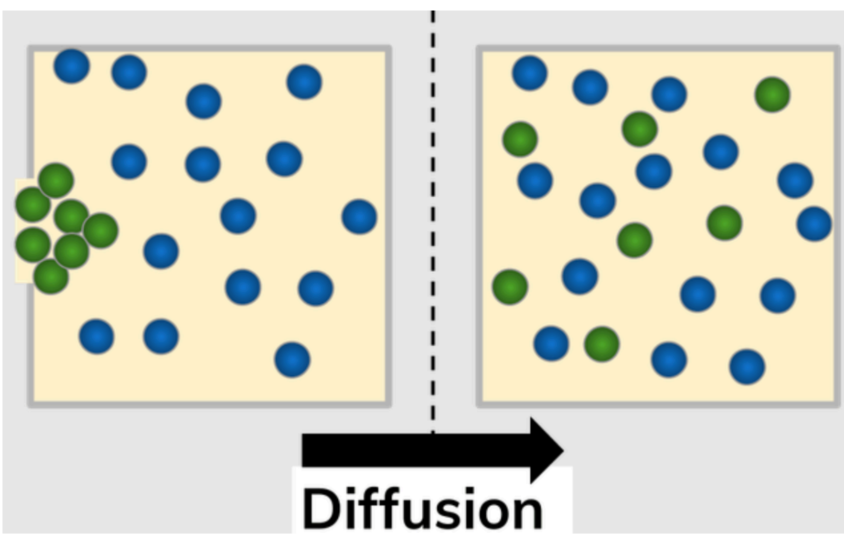
Benefits	Problems
Treat disease or injuries	Destroying embryos
Use unwanted embryos	Ethical/religious concerns
Save rare plants	Hard to control differentiation

🧪 7. Transport in Cells

Diffusion

Particles move from high to low concentration. No energy needed.

Examples: - Oxygen into blood - Carbon dioxide out - Glucose into cells



Affected By:

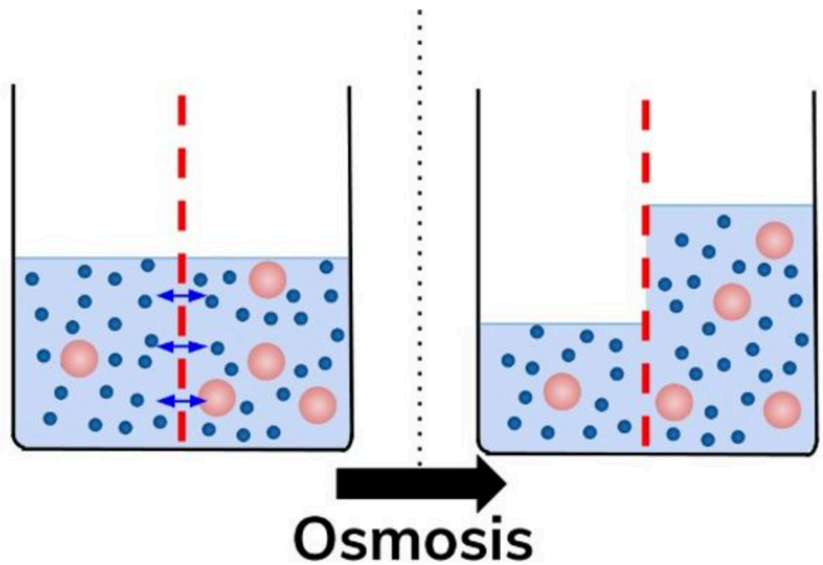
Factor	Effect
Temperature	Higher temp = faster diffusion

Factor	Effect
Surface area	More area = faster diffusion
Concentration	Bigger difference = faster rate

Osmosis

Water moves from high to low water concentration through a partially permeable membrane.

- Into animal cell: may burst
- Out of animal cell: may shrink
- Into plant cell: becomes firm (turgid)
- Out of plant cell: shrinks (plasmolysis)



Isotonic solution: No net (total) change
Hypertonic: Water moves out
Hypotonic: Water moves in

Active Transport

Particles move from low to high concentration. Needs energy.

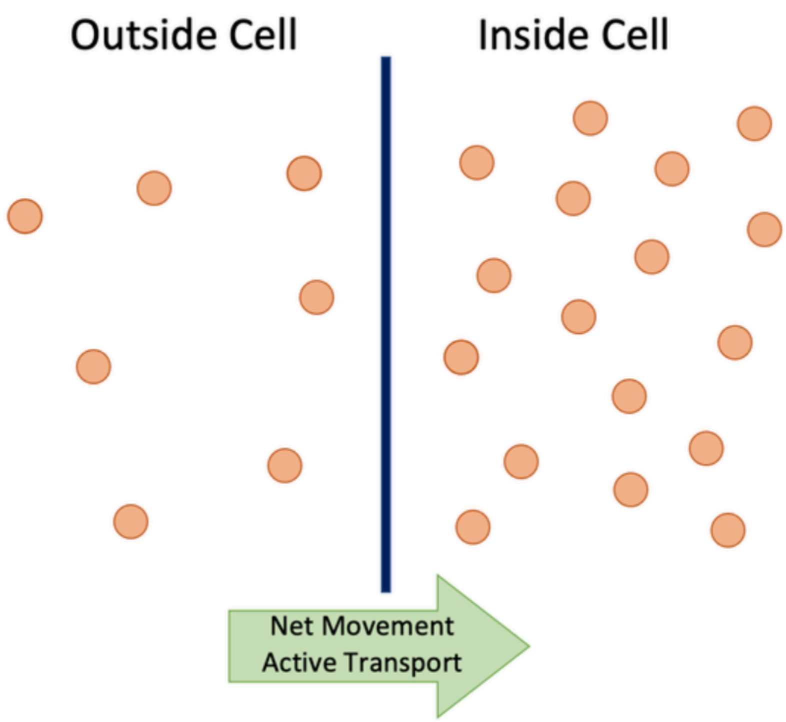
- In roots: minerals move in from soil
- In gut: sugar moves from gut to blood

🔄 Summary

Cell Biology is a foundation for all of biology. Learn the parts of the cell, how they function, how they move materials, and how they reproduce.

Remember: - Use clear, short sentences - Understand key terms - Practice calculations - Review each section with examples

You can do this!



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